California Polytechnic State University Pomona
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Circuit Analysis 2 Laboratory

ECE 2101L-03
Report #9

LAB 9 – PSpice Simulation

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Presented to
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April 5, 2024

Introduction:

In this lab we will be using PSpice, a circuit simulation tool, to analyze and verify different types of circuit setups. We will be simulating the following scenarios: Transient Analysis of a Switched RC Circuit, Transient Analysis of a Switched RLC Circuit, Transient Analysis of a Circuit with an AC Voltage Source, Transient Analysis of an RC Circuit Switch, Steady-State AC Analysis of an RC Circuit, AC Sweep of an RC Circuit, Using Plot Window Templates - Gain and Phase, Noise Analysis - Plotting Voltage Spectral Density, Second-Order Low-Pass Filter Gain, and Parametric Sweep. All of these will help us grow our familiarity with how to use PSpice and how to effectively analyze different circuits.

Intro Simulation:

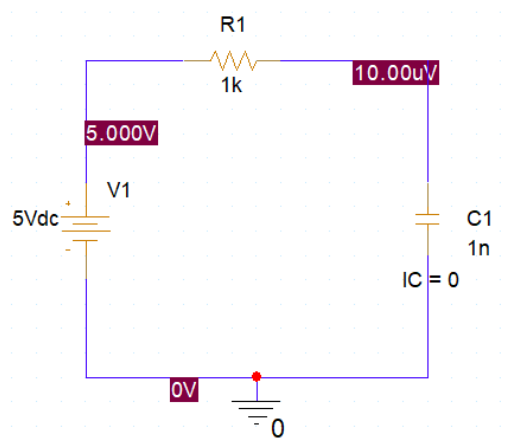


Figure 1: Intro Circuit

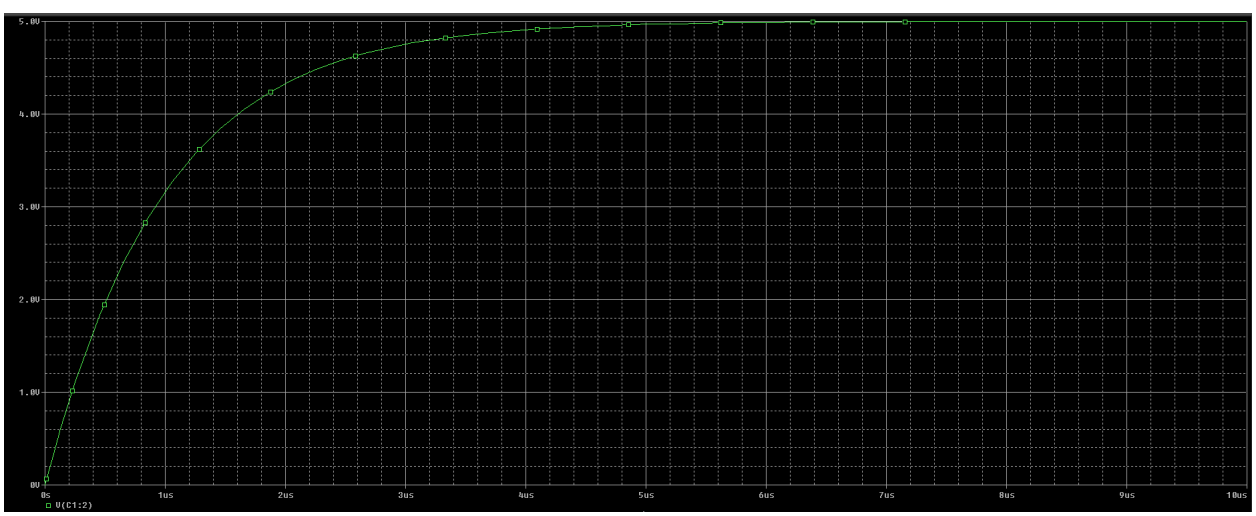


Figure 2: Voltage of Capacitor C1 from 0 to 10us

Transient Analysis of a Switched RC Circuit:

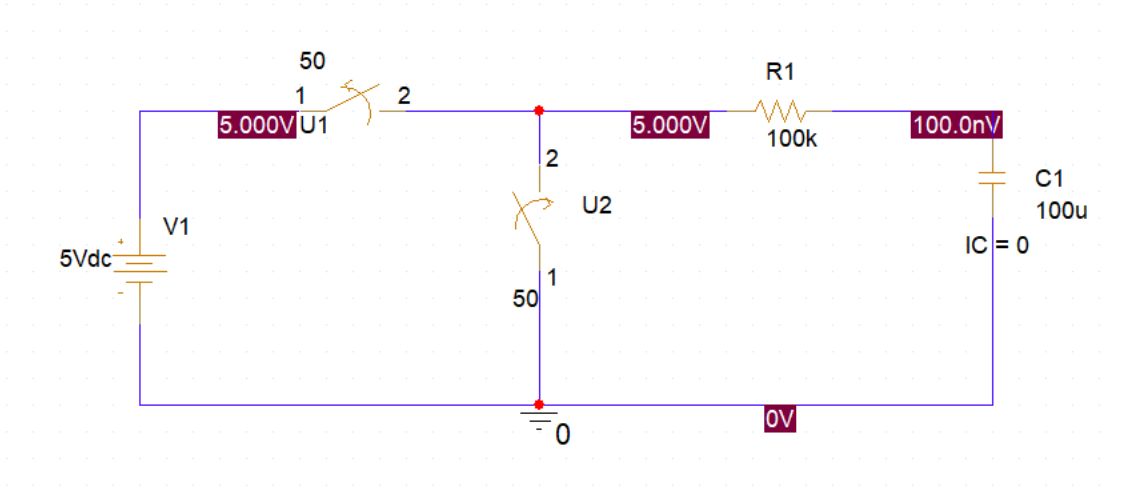


Figure 3: Transient Analysis of RC circuit

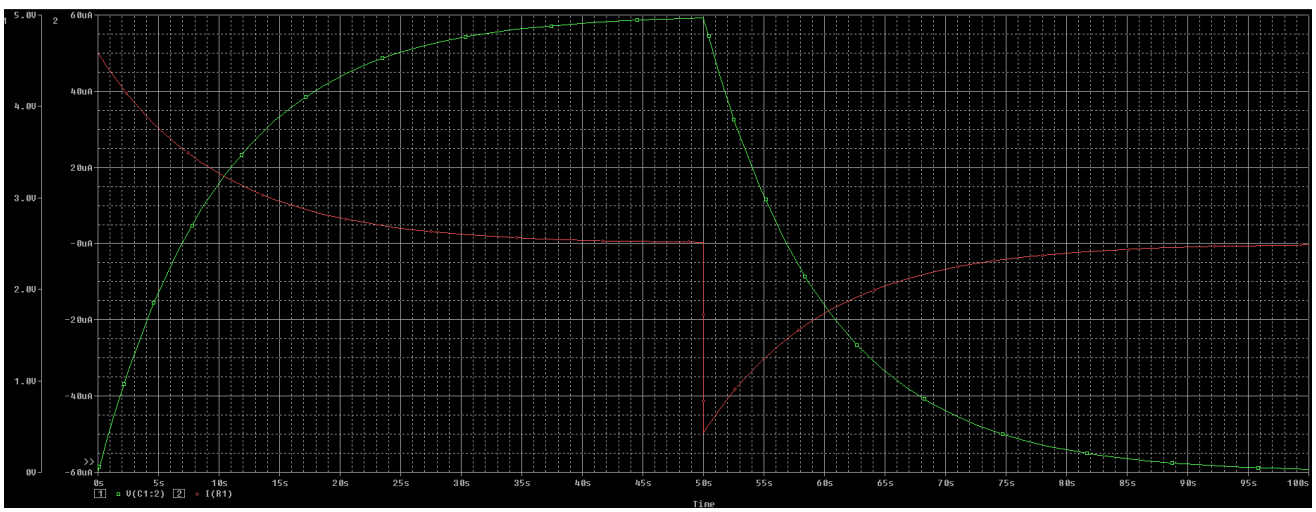


Figure 4: Voltage of Capacitor C1 (green) and Circuit Current Measured at the Resistor (red)

Transient Analysis of a Switched RLC Circuit:

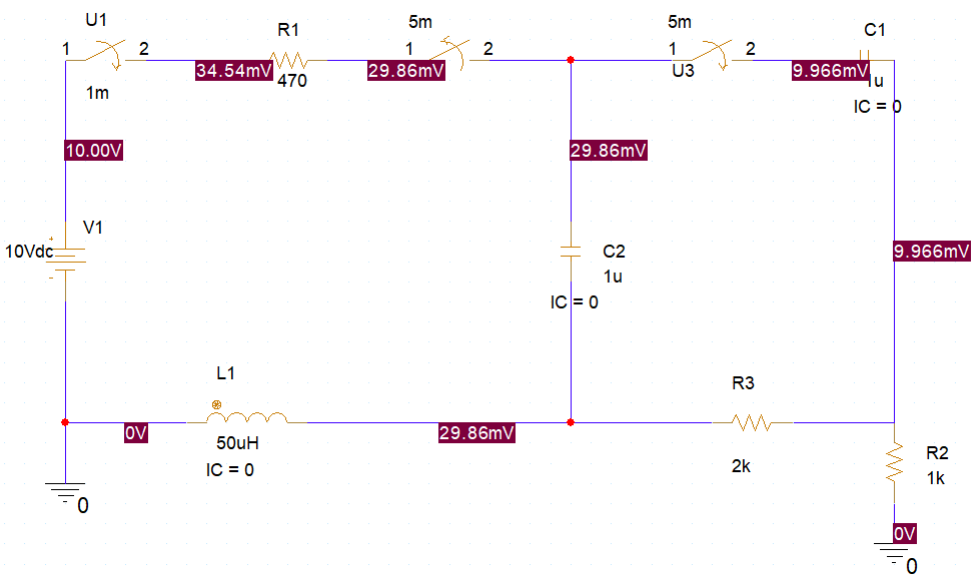


Figure 5: Transient Analysis of Switched RLC Circuit

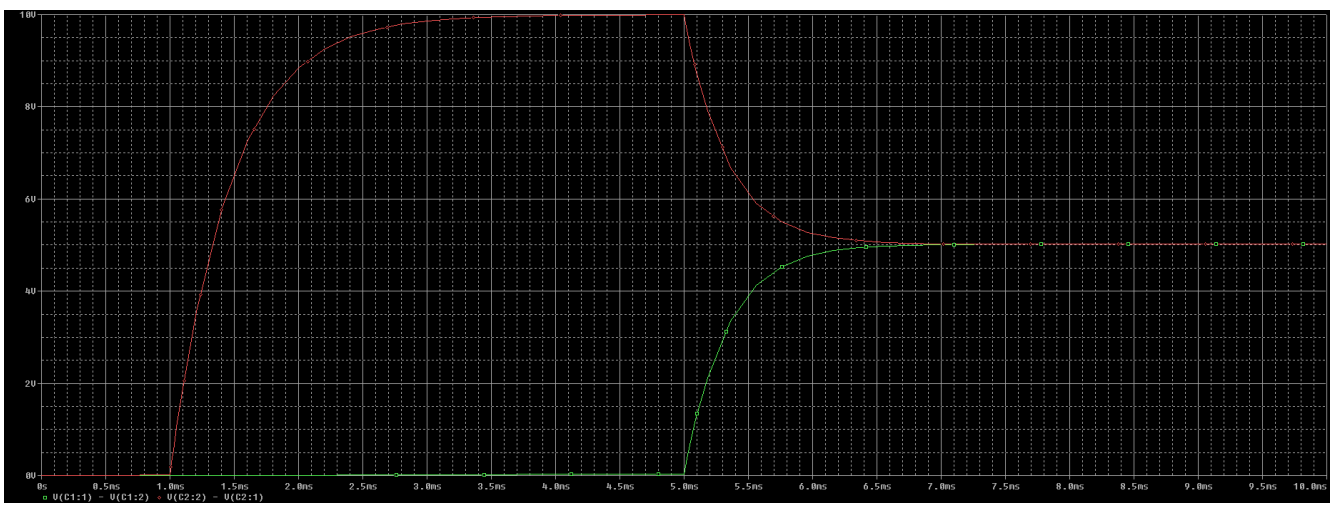


Figure 6: Voltage of Capacitor C2 (red) and Voltage of Capacitor C1 (green)

Transient Analysis of a Circuit with an AC Voltage Source:

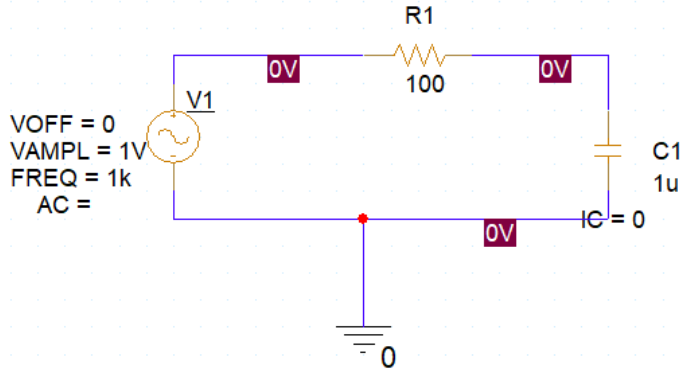


Figure 7: Transient Analysis of RC Circuit with AC Sine Source

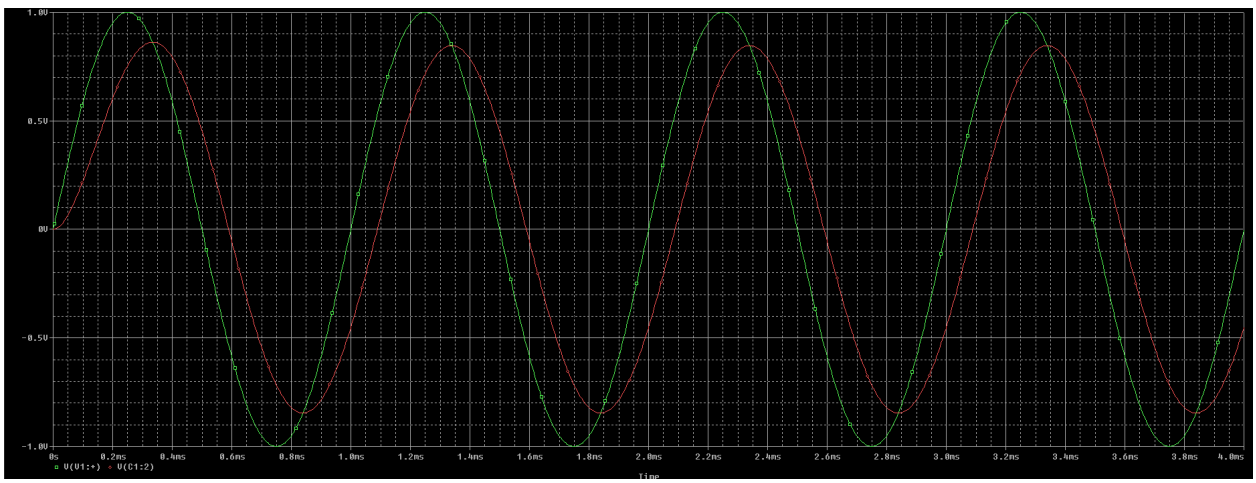


Figure 8: Voltage Source (green) and Voltage of Capacitor C1 (red)

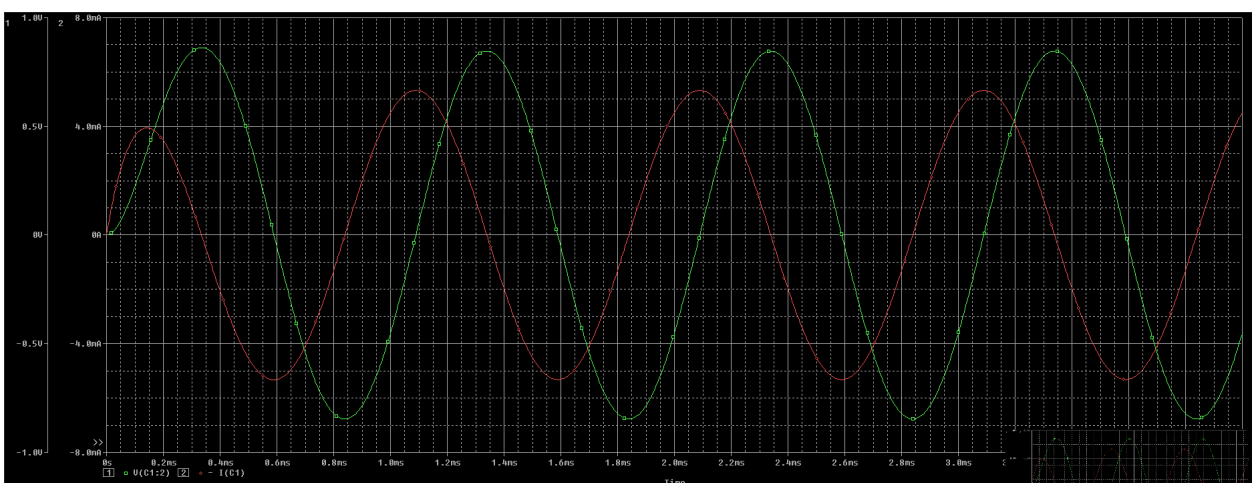


Figure 9: Voltage of Capacitor C1 (green) and Current of Capacitor C1 (red)

Exercise: Transient Analysis of a RC Circuit Switch

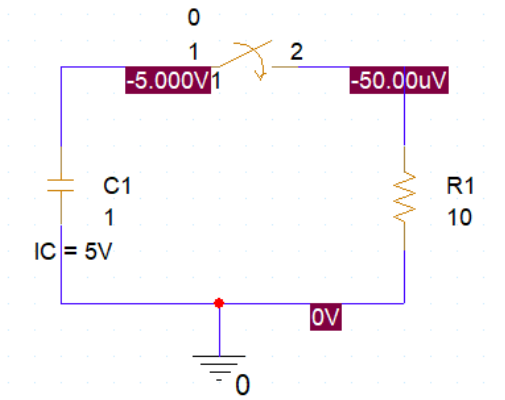


Figure 10: Transient Analysis of Sourceless RC Circuit

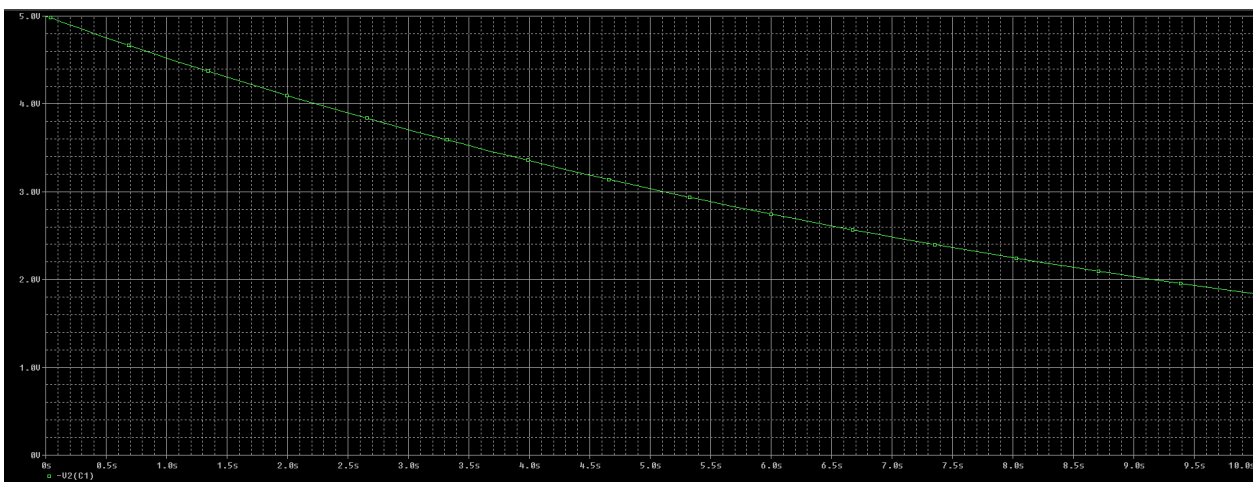


Figure 11: Voltage of Capacitor C1 (green)

Vtau = V0 * e^(-t/RC)

Vtau = 5 * e^(-10/10) = 1.839V

AC Analysis

Steady-State AC Analysis of a RC Circuit

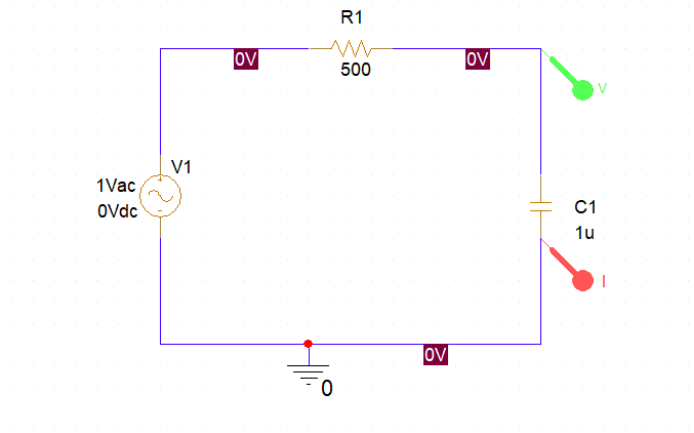


Figure 12: Steady-State AC Analysis of RC Circuit

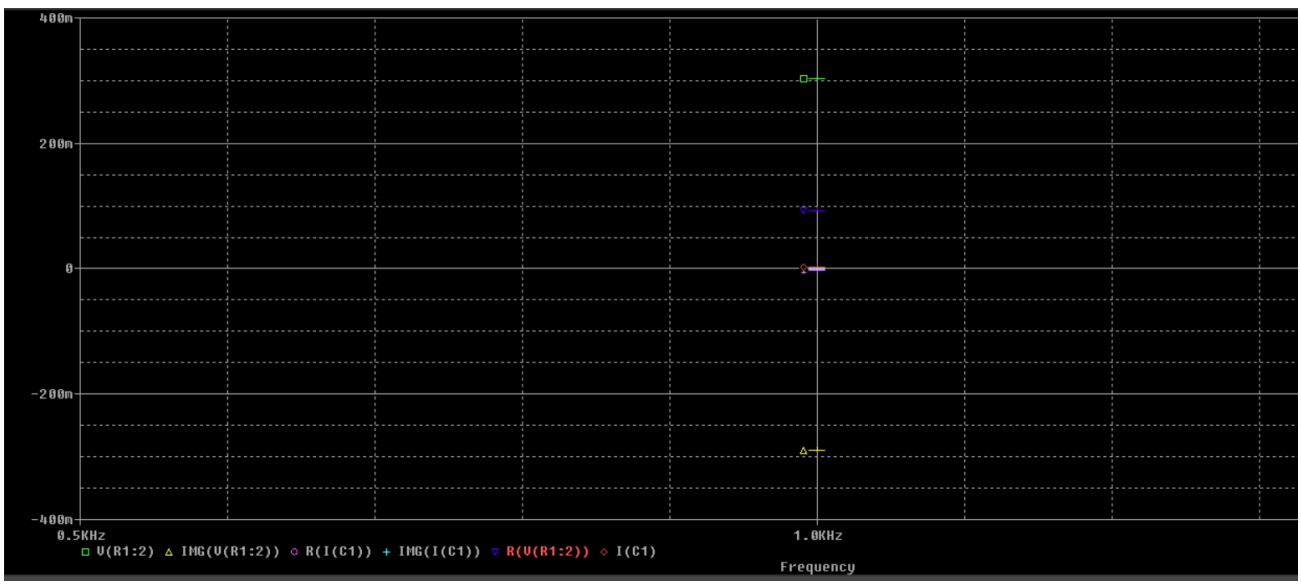


Figure 13: Data points of Voltages and Currents (Real and Imaginary)

Frequency	V(R1:2)	R(V(R1:2)	IMG(V(R1:2))	I(C1)	R(I(C1))	IMG(I(C1))
1.000000000000e+03	3.033144710534e-01	9.199966835038e-02	-2.890254822222e-01	1.905781027977e-03	-1.816000663299e-03	-5.780509644445e-04

Table 1: Voltages and Current Data Points Printed

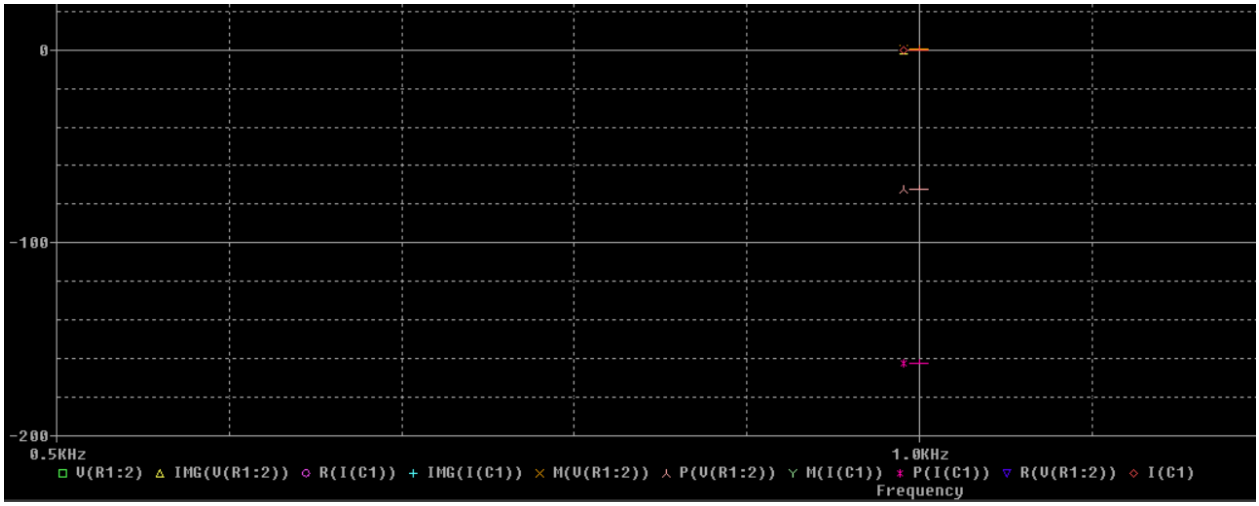


Figure 14: Data points of Voltages and Currents (Real, Imaginary, Phase and Magnitude)

M(V(R1:2))	P(V(R1:2))	M(I(C1))	P(I(C1))
3.033144710534e-01	-7.234321177987e+01	1.905781027977e-03	1.765678689057e+01

Table 2: Voltages and Current Data Points Printed

AC Sweep of a RC Circuit

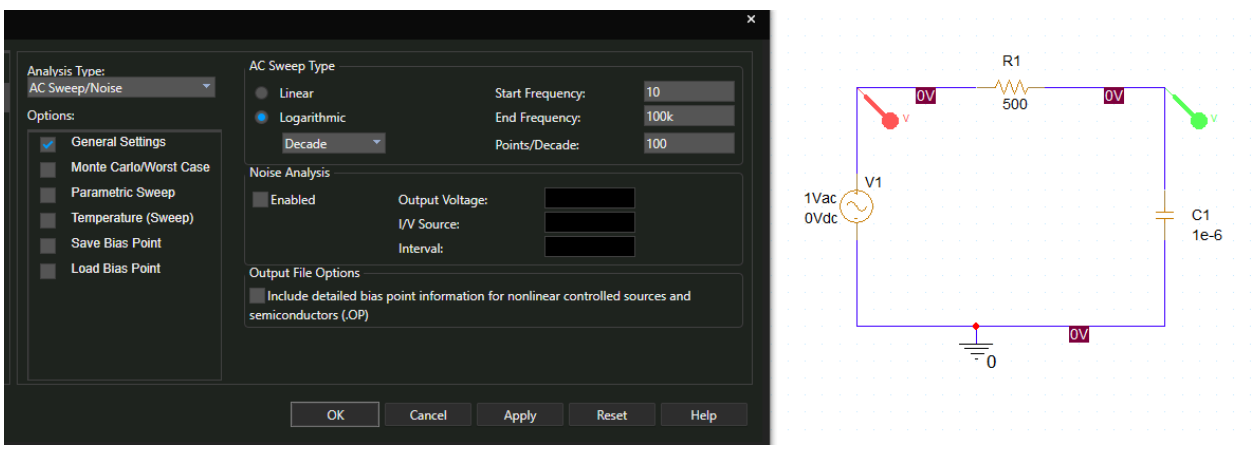


Figure 15: Pspsice AC Sweep Simulation Settings and RC Circuit

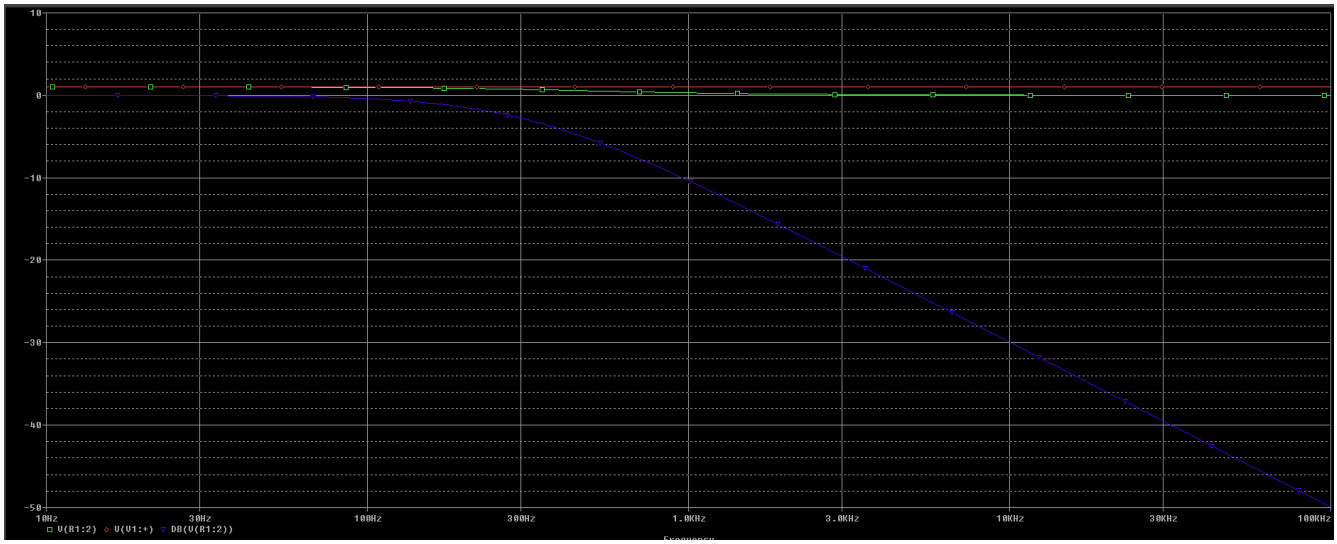


Figure 16: Voltage of R1 (green), Voltage of Capacitor (red) and DB of Output Voltage (blue)

Using Plot Window Templates - Gain and Phase

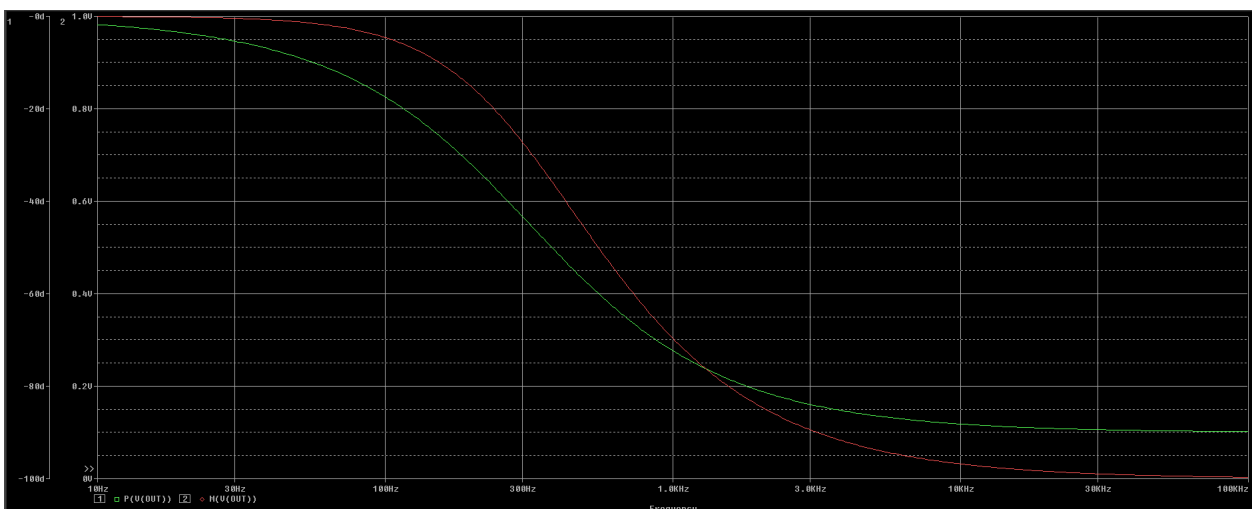


Figure 17: Phase (green) and Magnitude (red) of Output Voltage

Noise Analysis - Plotting Voltage Spectral Density

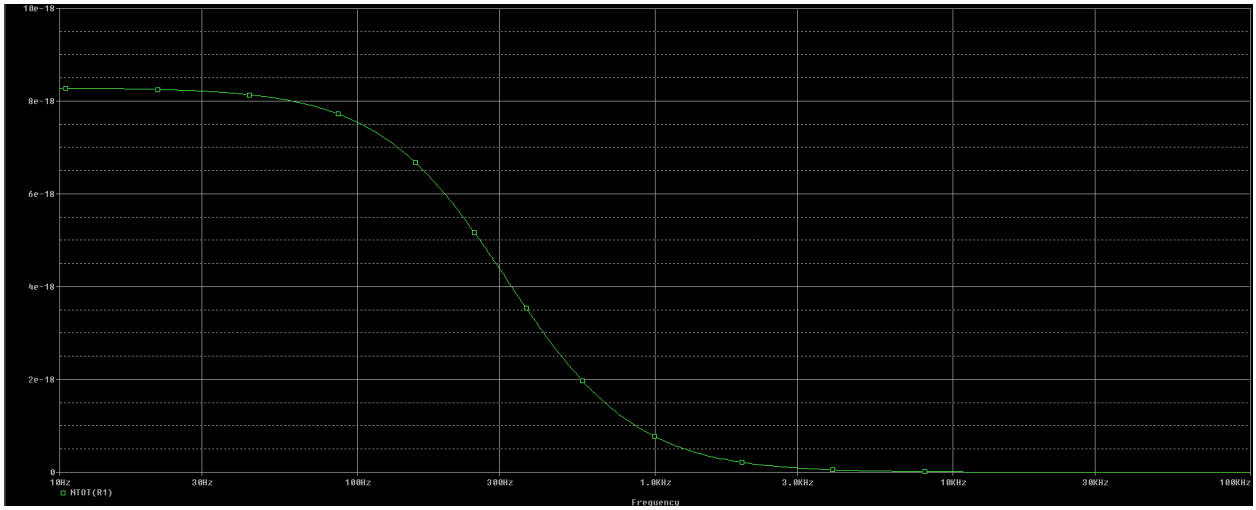


Figure 18: Voltage Spectral Density for Resistor R1

Second-Order Low-Pass Filter Gain

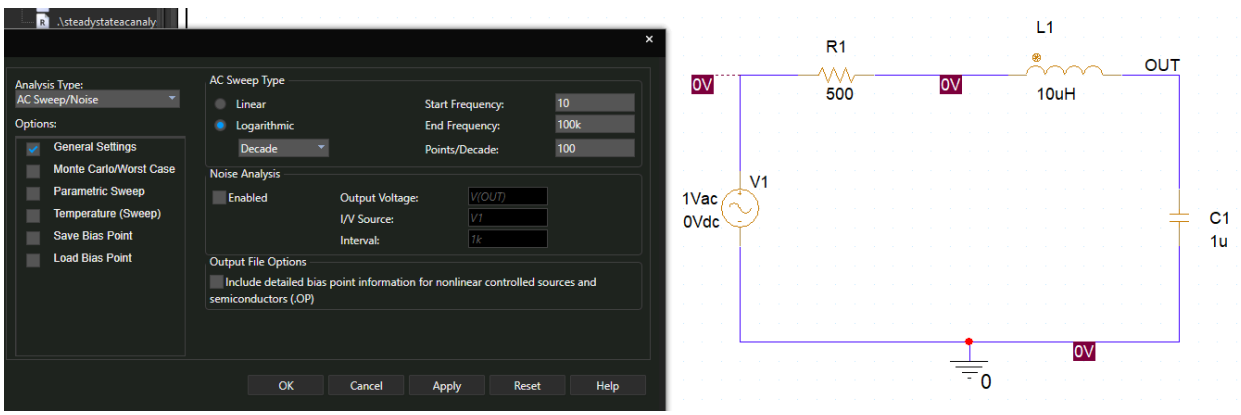


Figure 19: AC Sweep Settings and LPF RLC Circuit

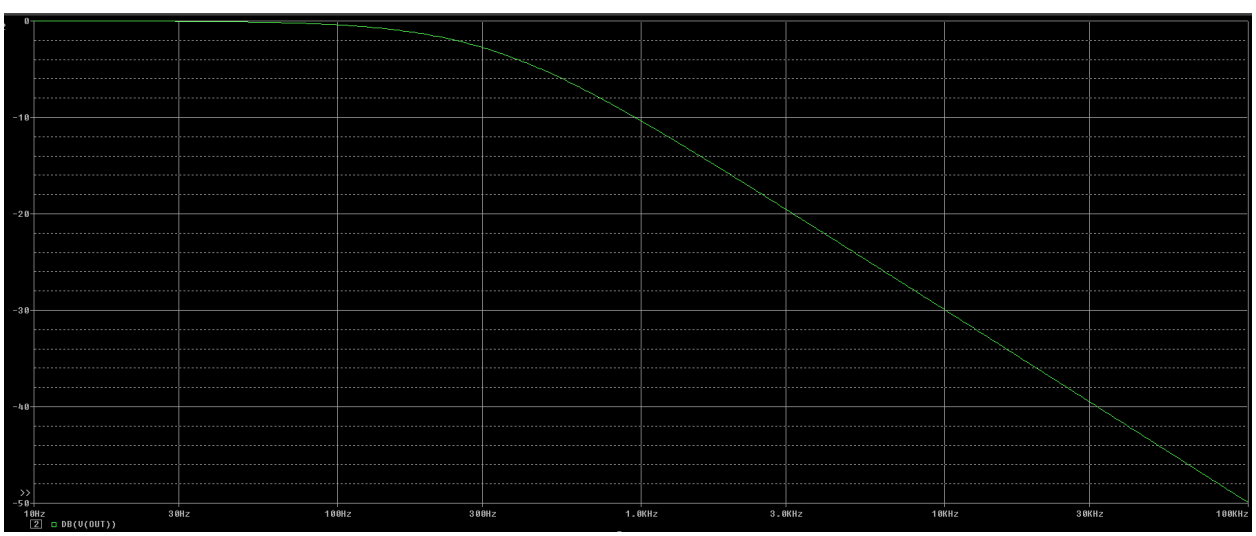


Figure 20: Gain of LPF RLC Circuit

Parametric Sweep

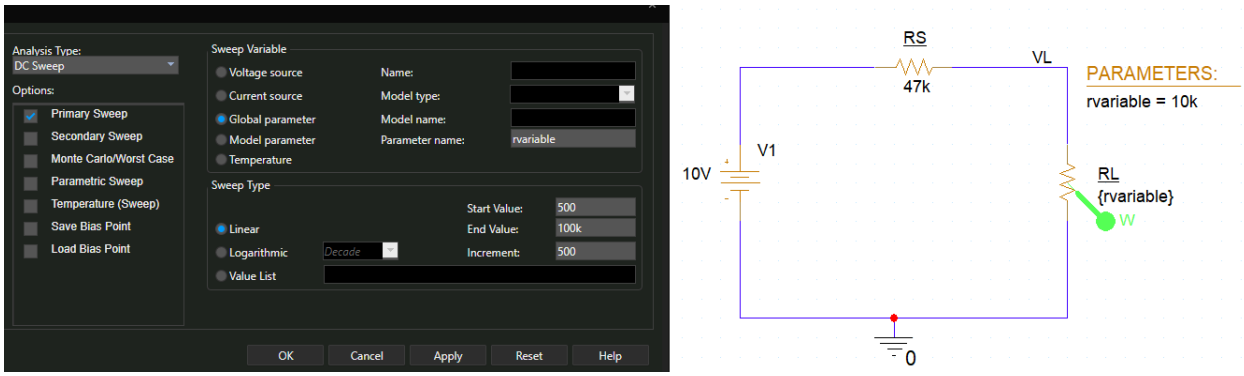


Figure 21: Parametric Sweep Settings and Circuit diagram

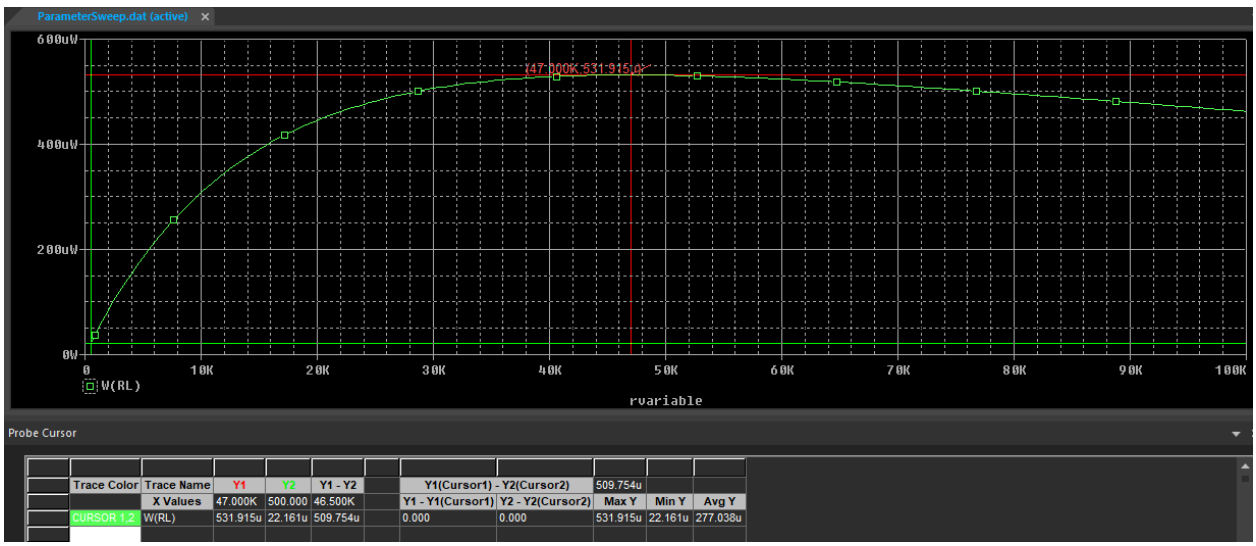


Figure 22: Power Dissipation of R1 (green) showing the Peak Power at 47k Ohms (red)

Post-Labs:

1.

Figure 23: RC Circuit with DC Source

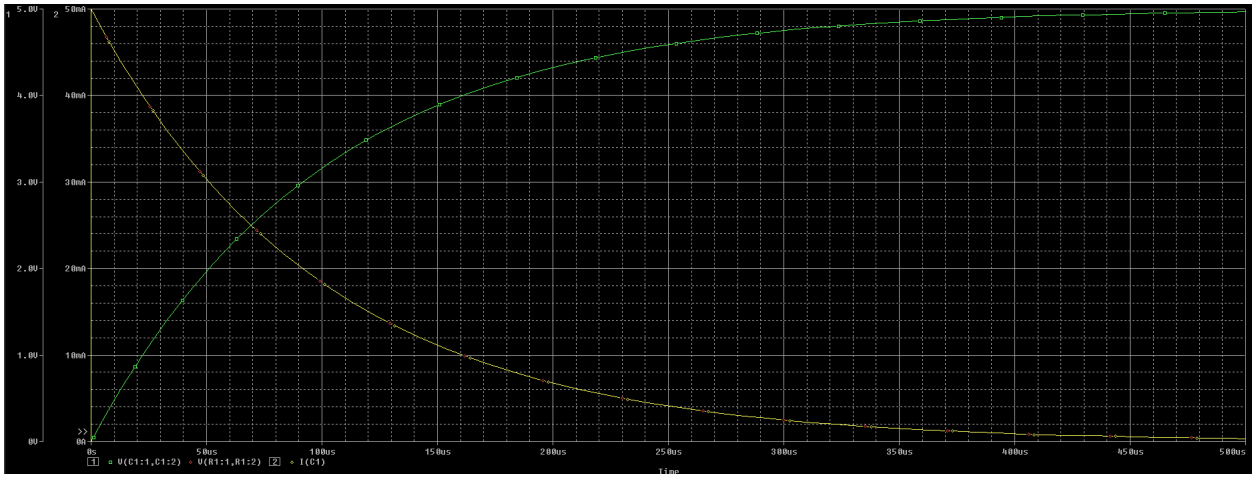


Figure 24: Voltage Across Capacitor C1

5*tau = 5*(RC) = 5*(100*1*10^-6)=0.0005

Vc = 1.84V

2.

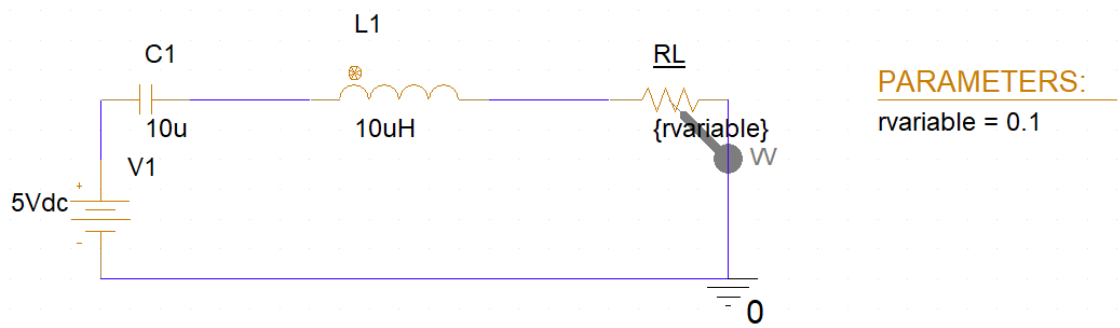
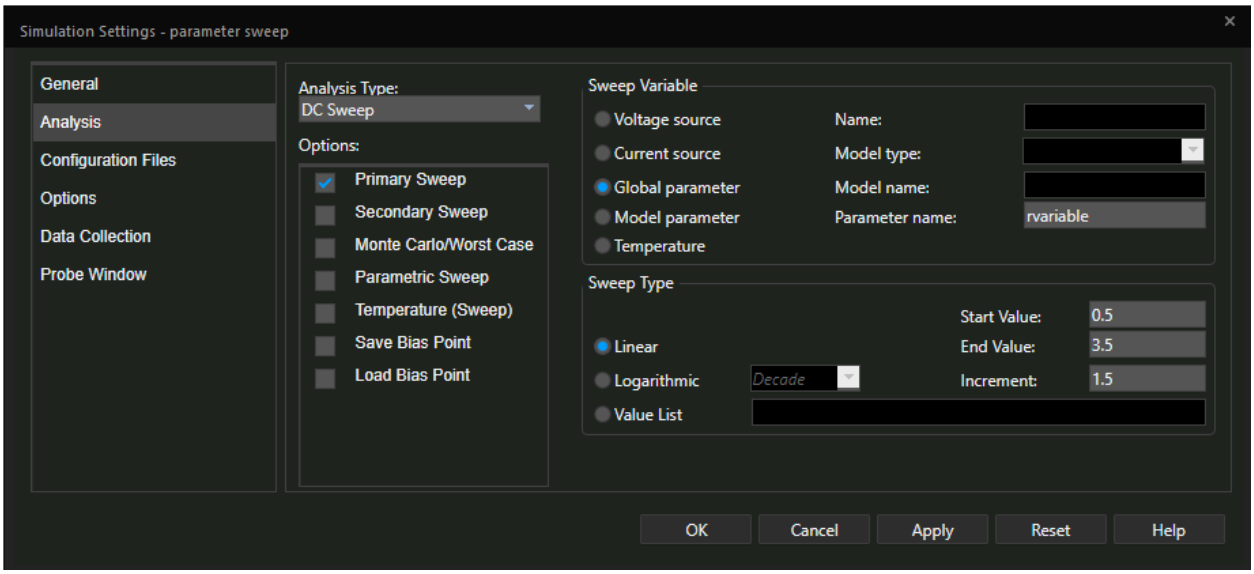


Figure 25: RLC Circuit with Variable Resistance



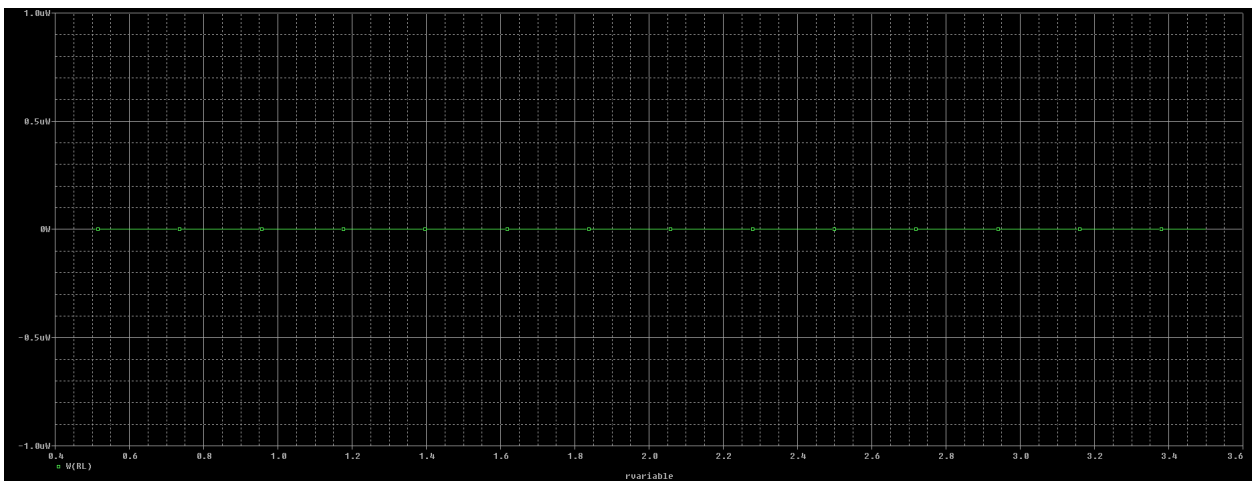
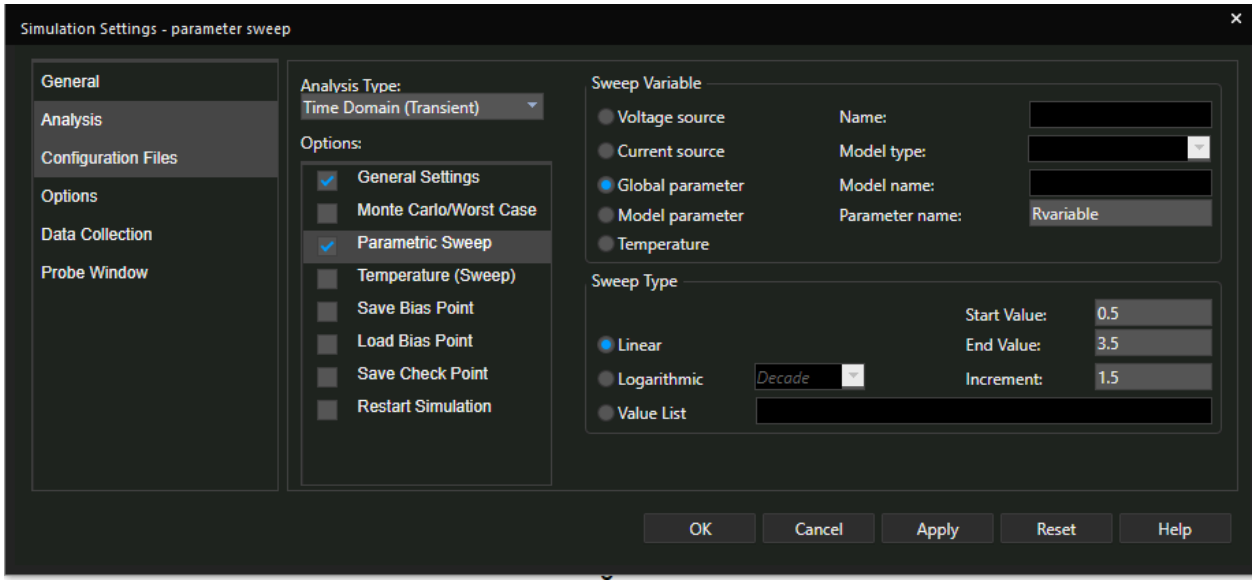


Figure 26 & 27: DC Sweep Simulation Settings and Power Graph



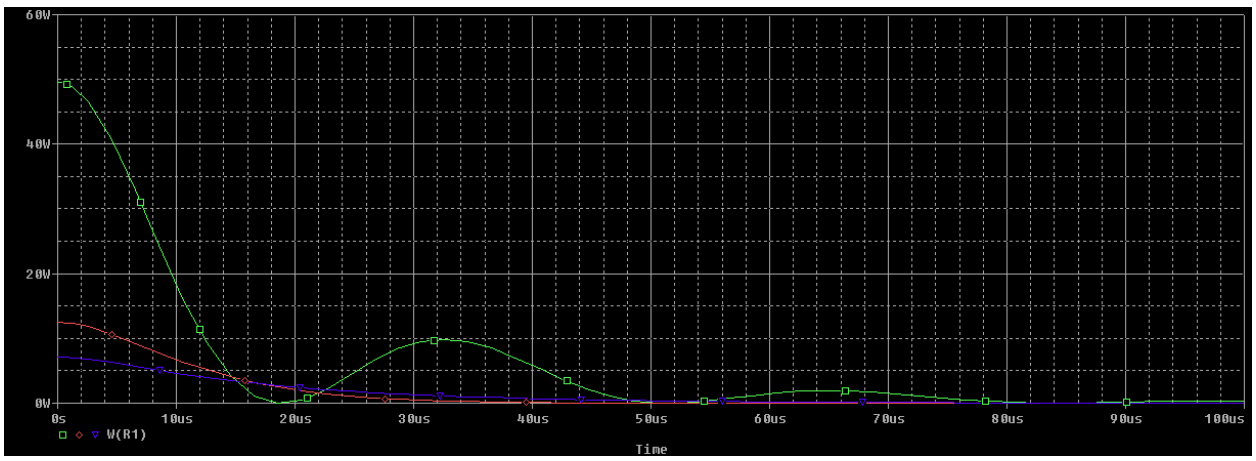


Figure 28 & 29: Transient Parametric Sweep Settings and Power Graphs

0.5 Ohms (green), 2 Ohms (red) and 3.5 Ohms (blue)

3.

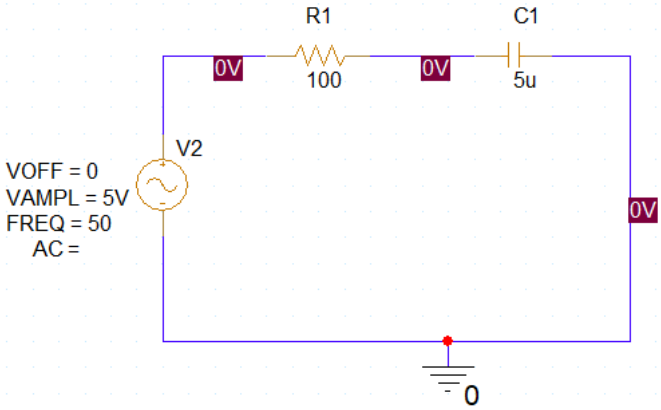


Figure 30: RC Circuit with AC Sine Source

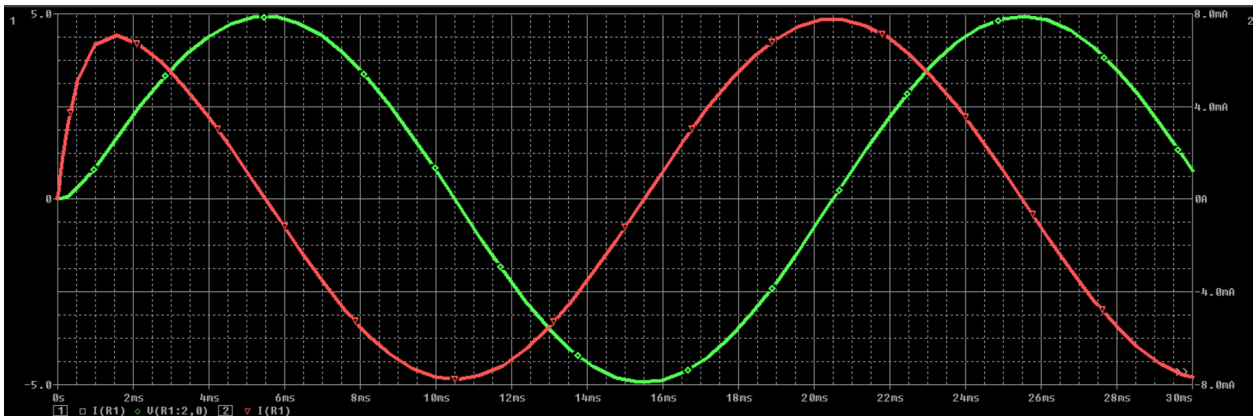


Figure 31: Voltage Across Capacitor C1 (green) and Current of Resistor R1 (red)

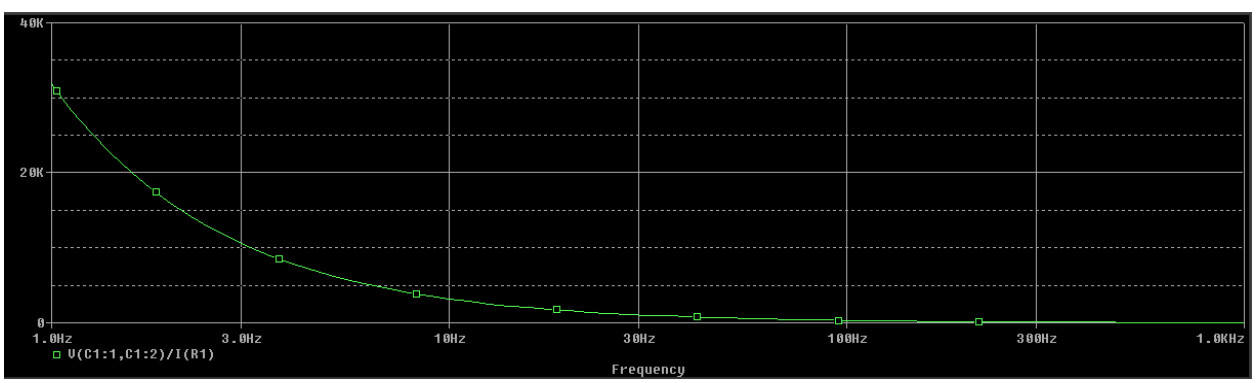


Figure 32: Reactance of the Capacitor in AC Sweep

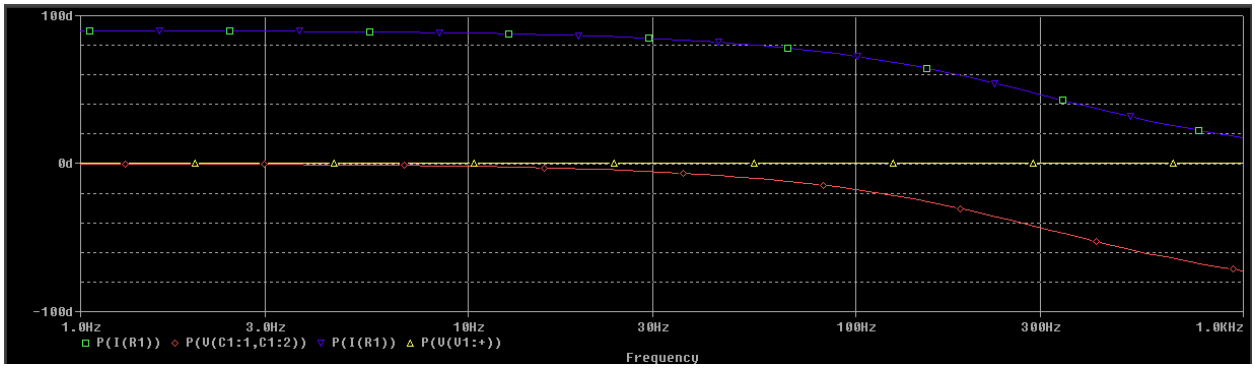


Figure 33: Phase of the Voltages Currents in AC Sweep

4.

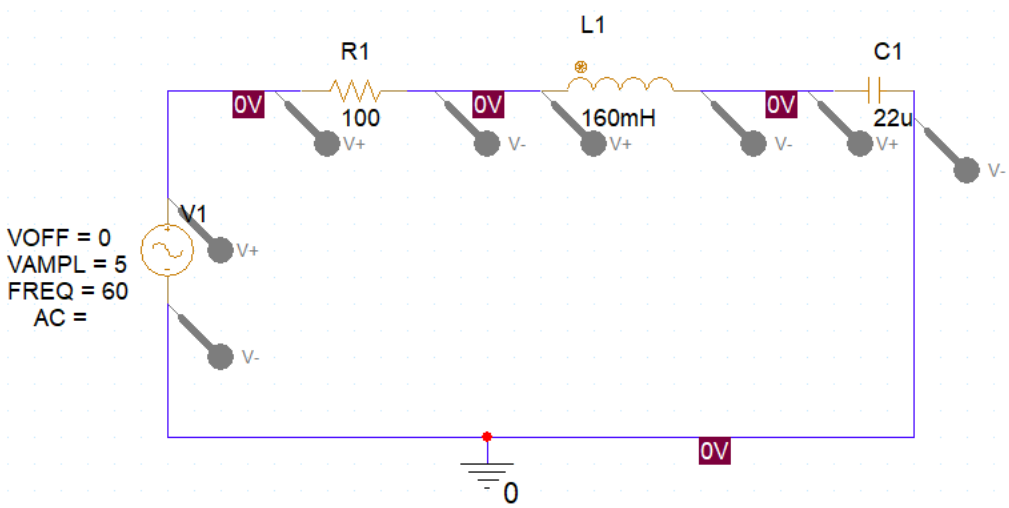


Figure 34: RLC Circuit with AC Sine Source

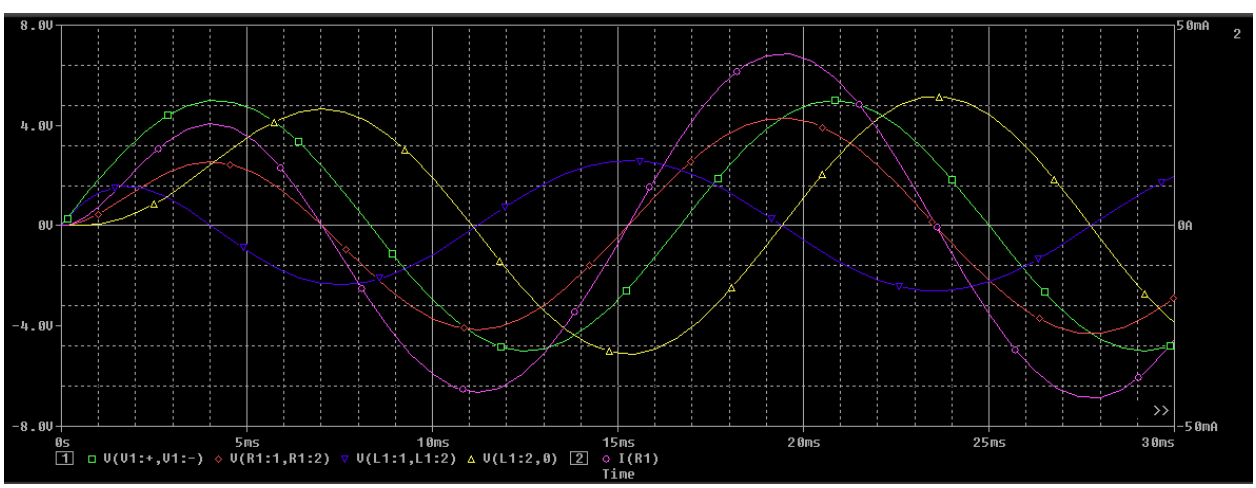


Figure 35: Voltages Across Source, Resistor, Inductor, Capacitor and Total Current (right Axis)

Conclusion:

Throughout this lab, we simulated and analyzed many different circuit scenarios such as Transient, AC, DC, and Parametric Sweep. We began with the basics, like where to find parts, how to place them, and how to change them. values. Then we moved on to the simulation portion, and how to change the Analysis type and parameters. Overall, this lab was very informative since it familiarized us with how to use PSpice effectively, which is important since PSpice is the industry standard.